

Beck's attempt

Solution: $f(x) = \begin{cases} 3x+1 & \text{when } x \geq 0 \\ -3x+2 & \text{when } x < 0 \end{cases}$

If $x \geq 0$, $\therefore 3x \geq 3 \cdot 0 \Rightarrow 3x \geq 0 \Rightarrow 3x+1 \geq 0+1 \Rightarrow 3x+1 \geq 1 > 0$.
 If $x < 0$, $\therefore -3x > -3 \cdot 0 \Rightarrow -3x > 0 \Rightarrow -3x+2 > 2 > 0$.

Let, $\exists a, b \in \mathbb{Z}$ s.t. $f(a) = f(b)$

Case 1: $a \geq 0$ and $b \geq 0$. $\therefore 3a+1 = 3b+1 \Rightarrow 3a = 3b \Rightarrow a = b$.

Case 2: $a \geq 0$ and $b < 0$. $\therefore 3a+1 = -3b+2 \Rightarrow 3(a+b) = 1 \Rightarrow a+b = \frac{1}{3}$.

$a, b \in \mathbb{Z}$. $\therefore$ This is impossible.

Case 3: $a < 0$ and $b \geq 0$. $\therefore -3a+2 = 3b+1 \Rightarrow 3(a+b) = 1 \Rightarrow a+b = \frac{1}{3}$.

Similar to Case 2, this is impossible.

Case 4: $a < 0$ and $b < 0$. $\therefore -3a+2 = -3b+2 \Rightarrow -3a = -3b \Rightarrow a = b$.

By Defn. 5, f is one-to-one.

~~Let $f(x) = 3 \in \mathbb{Z}^+$~~ Let, $f(x) = 3 \in \mathbb{Z}^+$

If $x \geq 0$, $\therefore 3x+1 = 3 \Rightarrow x = \frac{2}{3} \notin \mathbb{Z}$

If $x < 0$, $\therefore -3x+2 = 3 \Rightarrow x = -\frac{1}{3} \notin \mathbb{Z}$.

$\therefore \forall x \in \mathbb{Z}, f(x) \neq 3$. By Defn. 7, f is not onto. $\square$

b) onto, but not one-to-one.

Solution: $f(x) = |x|+1$

We know, $|x| \geq 0$. $\therefore |x|+1 \geq 1 > 0 \Rightarrow |x|+1 \geq 1 > 0$

$f(-1) = |-1|+1 = 1+1 = 2$

$f(1) = |1|+1 = 1+1 = 2$

But $1 \neq -1$. By Defn. 5, f is not one-to-one.

Let, $f(x) = y$ where $y \in \mathbb{Z}^+$. $\therefore |x|+1 = y \Rightarrow |x| = y-1$

If $x \geq 0$, then $|x| = x = y-1$

If $x < 0$, then $|x| = -x = y-1 \Rightarrow x = 1-y$. It's easy to see that both

$y-1 \in \mathbb{Z}$ and $1-y \in \mathbb{Z}$. $\therefore \forall y \in \mathbb{Z}^+, \exists x \in \mathbb{Z}$ s.t. $f(x) = y$.

By Defn. 7, f is onto. $\square$